

A series of overlapping circles in a light gray color, spanning the width of the page. On the far left, a circle is partially cut off by the edge. On the right, four circles overlap in a chain, with the last one also partially cut off.

MODUL IV

Image Morphology

From:
2026 Digital Image Lab Assistant



To:
2026 Digital Image Processing Practicum
Participants

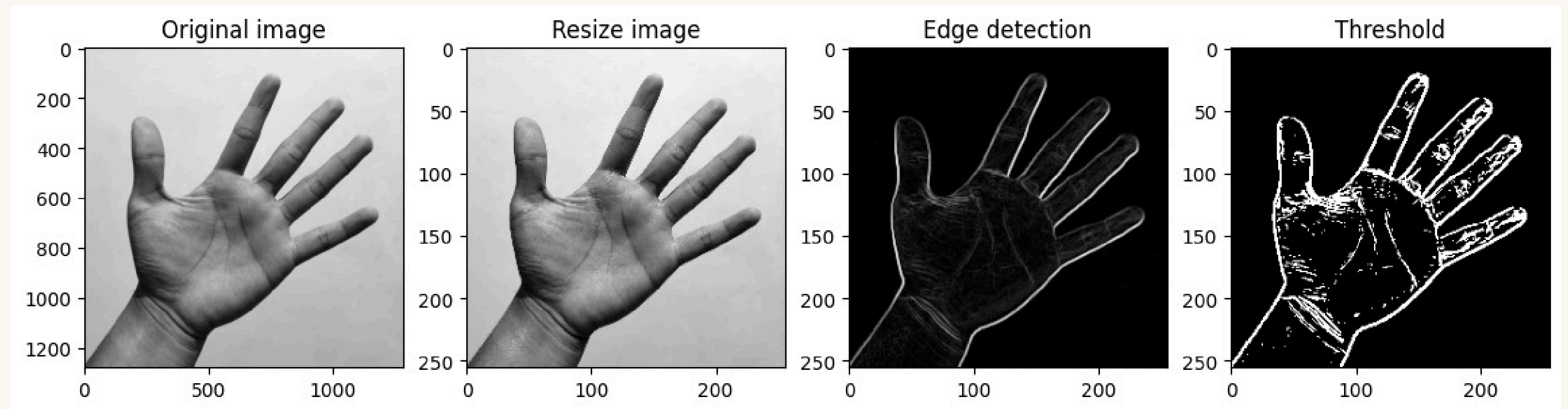


Image Preparation

Before applying morphological operations, perform image preprocessing first. Resize the image to 256 x 256 pixels, perform edge detection, and apply a thresholding technique to produce a binary image (with values of 0 or 1).



Result:



3x3 Kernel Image Morphological Operation

Dilasi



Erosi



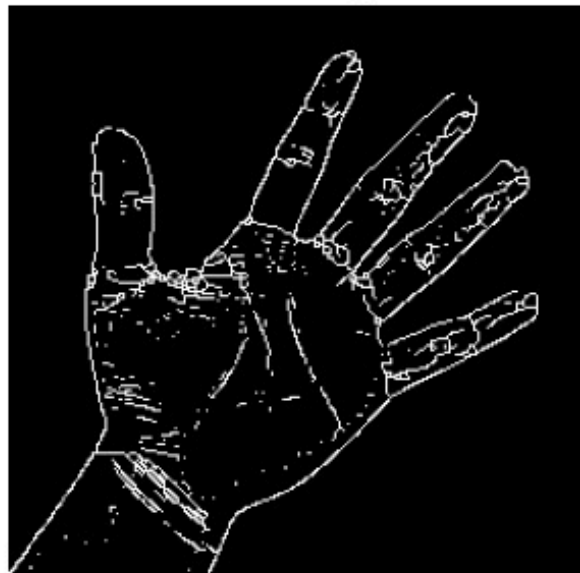
Opening



Closing



Thinning



Thickening



Gradient Morph



Morphology of a 3x3 Kernel

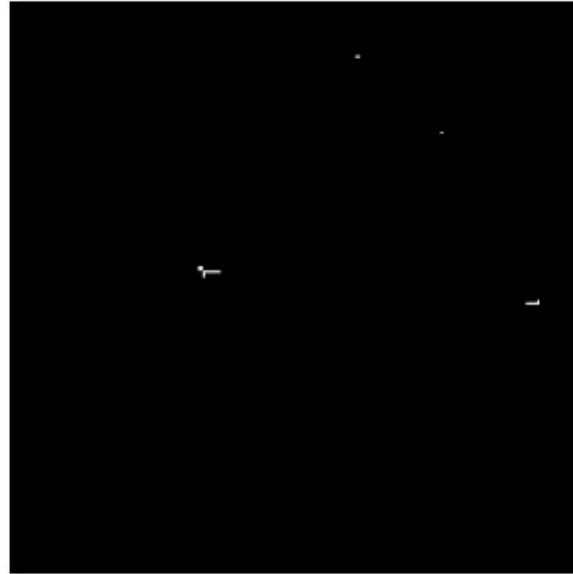
Perform all image morphological operations using a 3×3 kernel whose elements all have a value of 1. Display the results of each morphological operation, then analyze the visible differences between the morphologically processed image and the original image before the morphological process was performed!

5x5 Kernel Image Morphological Operation

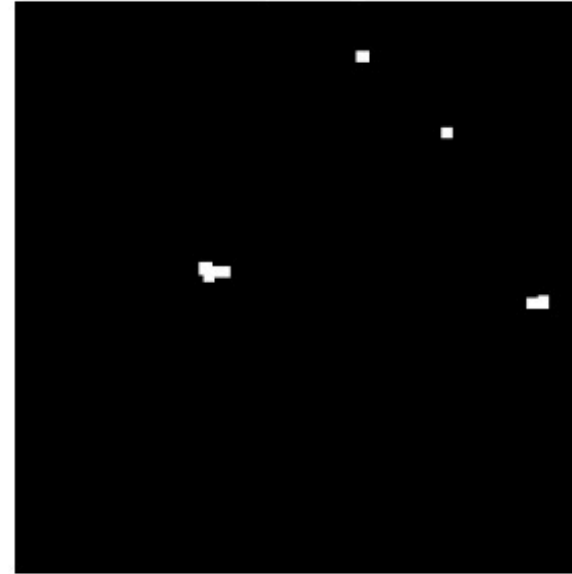
Dilasi



Erosi



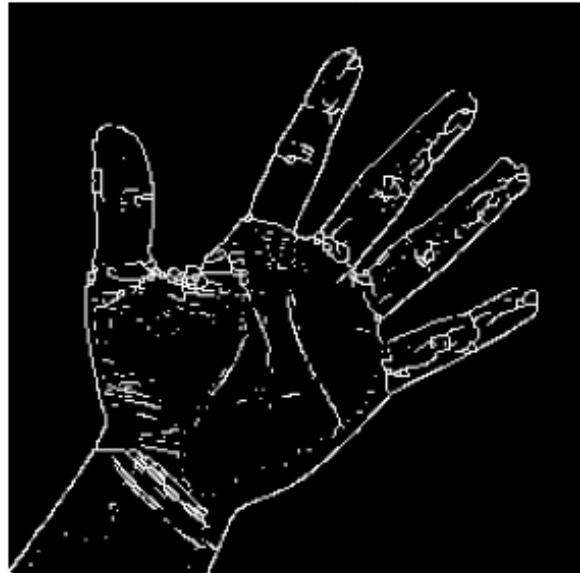
Opening



Closing



Thinning



Thickening



Gradient Morph



Morphology of a 5x5 Kernel

Perform all image morphological operations using a 5×5 kernel in which all elements have a value of 1. Display the results of each morphological operation, then analyze the visible differences between the images produced by the 5×5 kernel and the 3×3 kernel. Note the changes in object thickness, image detail, noise level, and object shape after the morphological processing is complete.

Combination of Two Morphological Processes

Perform a combination of two image morphological operations in sequence, such as opening–closing or erosion–dilation, using predefined kernels. Display the resulting images from each combination of operations, then analyze the changes that occur compared to the results of single morphological operations and the original image. Note the effect of the combination of operations on noise reduction, object shape, object thickness, and the clarity of object boundaries in the image.

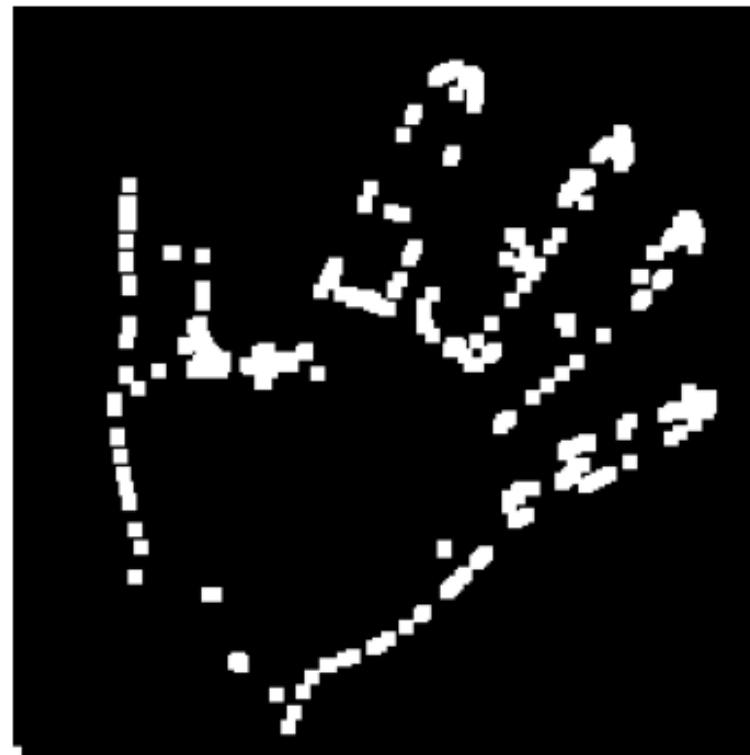
Opening + Closing



Closing + Opening



Opening + Dilasi



Closing + Erosi



Independent Exploration

Explore various image processing techniques on each student's palm image. Students are permitted to use any techniques they deem appropriate, such as image enhancement, filtering, thresholding, edge detection, resizing to various dimensions, and image morphological operations. Select and combine the most effective processes to produce a final image consisting of a clearly visible outline or contour of the palm. Display the results of each processing step, then provide an analysis of the reasons for using each process and its impact on the final image.

